KDS-EL: A Knowledge-Guided Evolutionary Learning Method for High-Dimensional Feature Selection

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Abstract

Feature selection in high-dimensional data, where the number of features vastly exceeds the number of samples, remains a central challenge for evolutionary computation. Generic particle swarm optimization (PSO) suffers from three structural deficiencies in this setting: random initialization wastes search budget in sparse solution spaces, uniform mutation discards cross-iteration feature-importance signals, and fixed objective weights are misaligned with the distinct phases of the search. This paper proposes KDS-EL, a knowledge-guided evolutionary learning method that builds upon a multi-task PSO backbone and integrates three complementary mechanisms. Mutual-information-guided initialization with adaptive re-initialization supplies informed starting points and escapes local optima on stagnation. A Knowledge-guided Elite Refinement (KER) strategy accumulates a cross-iteration feature-weight matrix governed by exponential decay and flips elite solutions toward historically effective features at zero additional classifier evaluations. A Two-Phase Weight Scheduling strategy partitions the search into an aggressive compression phase followed by an accuracy-refinement phase. On 20 high-dimensional benchmark datasets spanning gene expression, text, image, and speech domains, KDS-EL is compared against 13 competitors, including five single-objective PSO variants and five multi-objective feature selection methods. KDS-EL ranks first with a mean accuracy of 0.9574 and only 87.8 selected features on average, outperforming the strongest single-objective and multi-objective competitors by +3.97 and +4.42 percentage points, respectively, while delivering a 3.4× speedup over the MEL backbone. Wilcoxon signed-rank tests confirm superiority over MEL on 20/20 datasets, and Holm–Bonferroni correction establishes statistical significance against all 13 competitors. The component contribution analysis shows that the three mechanisms yield cumulative gains of +7.06 percentage points in accuracy and compress the feature count from 2,722 to 88.

Keywords: high-dimensional feature selection, particle swarm optimization, mutual information, knowledge-guided search, multi-task evolutionary learning, two-phase weight scheduling

1. Introduction

High-dimensional data now pervades science and engineering, from gene expression microarrays to text-mining and computer-vision representations. In these domains the number of measured features routinely reaches tens of thousands while labeled instances rarely exceed a few hundred, a regime denoted $p \gg n$. Under this condition, standard learning algorithms overfit, computational overhead escalates, and model interpretability degrades—the curse of dimensionality [1, 2]. Feature selection (FS), the task of identifying a compact subset of informative features before model training, defends against these effects by eliminating redundant and irrelevant variables [1, 2]. Despite decades of development, high-dimensional FS remains an open challenge: combinatorial growth of the search space precludes exhaustive enumeration, while pervasive feature correlations confound simple marginal screening.

Feature selection methods fall into three paradigms. Filter methods rank features by statistical scores such as mutual information or correlation, independent of any learning algorithm; they are efficient but neglect feature interdependencies and classifier-specific relevance. Wrapper methods evaluate candidate subsets by training a classifier, achieving higher accuracy at the cost of substantial search overhead [3]. Embedded methods integrate selection into the training objective itself, as with L_1-regularized regression, balancing speed and accuracy but remaining tied to model assumptions. Among wrapper approaches, evolutionary algorithms—including particle swarm optimization (PSO), genetic algorithms, and differential evolution—perform strongly on high-dimensional data because population-based stochastic search explores enormous non-convex spaces without restrictive prerequisites. Gene expression datasets, with small samples ($N \approx 30$–100) and tens of thousands of features, are the de facto benchmark for evolutionary FS: scarce labels demand parsimonious subsets, while dense correlation among co-regulated genes strains deterministic heuristics [1, 4]. An effective FS algorithm must therefore deliver high accuracy, aggressively compress the feature set, and finish within feasible runtime—objectives that frequently conflict under a fixed evaluation budget.

Standard PSO implementations [5, 6] were designed for continuous optimization and later adapted to binary spaces

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via sigmoid position transformations. We identify three structural mismatches between this generic architecture and the requirements of sparse, high-dimensional FS. First, random initialization scatters particles uniformly across the binary hypercube; in a D-dimensional space where fewer than 3% of features are ultimately selected, early evaluations are overwhelmingly consumed on dense, low-accuracy subsets before any meaningful signal is obtained. KDS-EL addresses this cold-start inefficiency with mutual information (MI)-guided initialization: features are pre-ranked by MI to the class label, and the initial population is seeded with informed starting points that bias search toward a sparse, high-potential region. Second, uniform random bit-flips treat every dimension identically regardless of whether a feature has historically helped or hurt accuracy—the per-feature track record accumulated during search is discarded at each velocity update. KDS-EL resolves this knowledge-blind mutation via the Knowledge-guided Elite Refinement (KER) strategy: a cumulative feature-weight matrix, updated by fitness improvements and damped by exponential decay, guides elite perturbations toward historically effective features at no additional classifier cost. Third, the dual-objective fitness function that balances classification error against subset cardinality typically employs a fixed weight vector throughout search, ignoring the temporal structure of FS—non-discriminative features are best removed in early aggressive sweeps, whereas accuracy refinement benefits from precision-oriented weighting later. KDS-EL eliminates this rigid scheduling with a Two-Phase Weight Scheduling strategy that partitions search into a compression-dominated early phase and an accuracy-dominated late phase. These three components operate at complementary levels of the search architecture and build upon the multi-task PSO backbone introduced in MEL [7], forming a transparent incremental construction: MEL → MEL+MI → KDS-EL$_{\text{KER}}$ → KDS-EL.

This paper makes the following contributions:

1. To overcome the cold-start inefficiency of random initialization in high-dimensional sparse search, we propose an MI-guided initialization scheme with adaptive re-initialization that seeds the initial population with a mixture of MI-informed and random individuals, concentrating search on a promising fraction of dimensions from the outset.
2. To exploit the cross-iteration feature-importance signals discarded by uniform mutation, we propose the Knowledge-guided Elite Refinement (KER) strategy, which accumulates a cross-iteration feature-weight matrix from fitness-improvement signals, applies exponential decay to discount obsolete information, and employs knowledge-biased elite perturbation (90% guided, 10% random) at zero additional evaluation cost.
3. To align objective weights with the temporal structure of feature selection, we propose a Two-Phase Weight Scheduling strategy that replaces static dual-objective weights with a dynamic schedule: the early phase applies stronger feature-reduction pressure, and the late phase shifts to accuracy-dominant weighting, yielding simultaneous gains in accuracy, feature sparsity, and runtime.
4. Systematic experiments on 20 high-dimensional benchmark datasets spanning 500 to 54,675 features, against 14 fixed-length methods covering single-objective, multi-objective, and MEL baselines [7], show KDS-EL ranks first with mean accuracy **0.9574** and **87.8** selected features on average—leading KDS-EL$_{\text{KER}}$ by +0.86 pp, the strongest single-objective competitor FWPSO by +3.97 pp, and the strongest multi-objective method W-MOPSO-FS by +4.42 pp, with a 3.4× speedup over MEL (51.3 s vs. 175.0 s) and Wilcoxon significance on 20/20 datasets.

The remainder of this paper is organized as follows. Section 2 surveys related work in feature selection, evolutionary computation, and multi-task learning. Section 3 reviews background on binary PSO and the MEL multi-task framework. Section 4 presents the KDS-EL algorithm in full. Section 5 describes the experimental configuration. Section 6 reports and analyzes the main results. Section 7 provides a component contribution analysis. Section 8 examines parameter sensitivity. Section 9 concludes the paper.

2. Related Work

2.1. Feature Selection Paradigms

Feature selection is a fundamental pre-processing step in high-dimensional data analysis, aiming to identify a minimal subset of relevant features that preserves or improves classification performance while reducing computational cost and enhancing interpretability [1, 2]. Methods fall broadly into three paradigms: filter, wrapper, and embedded.

Filter methods evaluate features independently of any learning algorithm, relying on intrinsic data properties such as variance, correlation, or eigenvector centrality [8, 9]. Representative techniques include the minimum Redundancy Maximum Relevance (mRMR) criterion [10] and the Fast Correlation-Based Filter (FCBF) [11]. These methods are computationally efficient and scale readily to high-dimensional settings, yet they disregard feature interactions and may retain redundant subsets under strong inter-feature correlation.

Wrapper methods treat feature selection as a search problem, evaluating candidate subsets by the predictive performance of a specific classifier [3]. Hybrid genetic algorithms [12] and other meta-heuristic wrappers can capture complex feature interdependencies, but their computational cost grows rapidly with dimensionality, as each candidate subset requires a full classifier training-and-evaluation cycle.

Embedded methods integrate feature selection into model training. Lasso and its bootstrapped variant Bolasso [13] enforce sparsity through L_1 regularization; gradient-boosting-based approaches such as FS-GBDT [14] leverage tree-split importance; and hybrid algorithms like ILRC [15] combine regularization with clustering. Embedded methods offer a favorable speed–accuracy trade-off, but their feature rankings are tied to specific model assumptions.

At high dimensionality, each paradigm faces distinct structural bottlenecks: filters are fast but oblivious to feature interactions, wrappers are accurate but face a search space that grows exponentially with dimension, and embedded methods depend on model-specific regularization biases. These complementary weaknesses motivate evolutionary computation, which decouples the search mechanism from the evaluation metric and navigates the exploration–exploitation trade-off through population-based global search.

2.2. Single-Objective Evolutionary Feature Selection

Evolutionary algorithms have been extensively applied to feature selection for their population-based global search and gradient-free nature. Early work adapted classical metaheuristics—binary PSO (BPSO) [16], ant colony optimization (ACO) [17], artificial bee colony (ABC) [18], and genetic algorithms with support vector machines (GA-SVM) [19]—to the binary subset selection problem, establishing the template of a population of candidate subsets evolving under selection, crossover, and mutation guided by a scalar fitness function.

Because standard evolutionary operators degrade at high dimensionality, specialized single-objective methods have been developed: self-adaptive PSO adjusting swarm size and mutation rate [20], progressively refined PSO alternating between coarse- and fine-grained search [21], and problem-specific heuristics—FWPSO [22], which biases velocity updates with per-feature importance; SaWDE [23] with adaptive weighting; IGPSO [24], which estimates relevance via a multi-layer perceptron; and PSO-CSM [25] integrating multiple evaluation criteria. Learning-based control and space decomposition are also explored: reinforcement-learning-aided PSO [26] and divide-and-conquer search [27]. These methods refine search operators but do not systematically exploit accumulated feature-importance knowledge harvested from the search itself.

2.3. Multi-Objective Evolutionary Feature Selection

Multi-objective formulations treat classification accuracy and feature count as competing objectives, producing a Pareto front of non-dominated solutions. Xue et al. [28] pioneered multi-objective PSO for feature selection, showing that jointly optimizing accuracy and subset size yields more diverse and informative solutions than scalarized approaches. Hu et al. [29] extended this framework with fuzzy cost functions for imbalanced classification.

The multi-objective evolutionary algorithm (MOEA) family provides several architectures for this problem. NSGA-II-FS [30] uses non-dominated sorting and crowding distance to maintain Pareto-front diversity. MOEA/D-FS [31] decomposes the problem into scalar subproblems solved cooperatively. MOPSO-FS [32] extends single-objective PSO with an external archive and Pareto-dominance-based leader selection. For high-dimensional data, Cheng et al. [33] proposed a variable-granularity search method (VGS-MOEA) that transitions from coarse to fine-grained search, and Hancer et al. [34] combined evolutionary search with filter-based ranking under both single- and multi-objective settings.

Despite their conceptual advantages, MOFS methods incur substantial overhead: maintaining an external archive, non-dominated sorting, and Pareto-front management all cost significant time (in our experiments, over 110 seconds on average). Moreover, under fixed-length binary encoding at high dimensionality, the Pareto front expands combinatorially and convergence slows, limiting scalability at gene-expression scale.

2.4. Evolutionary Multi-Task Learning

Evolutionary multi-task learning (EMTL) solves multiple related tasks jointly, with knowledge transfer accelerating convergence. Zhang and Yang [35] surveyed its foundations, and several instantiations target feature selection: a minion-based multitasking framework [36], PSO-based multitasking for high-dimensional classification [37, 38], multitask bee colony optimization for hyperspectral band selection [39], and multitasking for large-scale multi-objective optimization [40].

Of particular relevance, Wang et al. [7] proposed MEL (Multi-task Evolutionary Learning), a dual-swarm multi-task PSO in which one swarm optimizes binary feature subsets while a second learns continuous feature weights, with the swarms cooperating through cross-task knowledge transfer. MEL was selected as the backbone for three reasons: its fixed-length encoding is efficient at high dimensionality and naturally compatible with incremental augmentation; its dual-space cooperation provides a built-in mechanism for capturing feature relevance signals; and it has demonstrated strong performance on gene-expression benchmarks.

Nevertheless, MEL exhibits three structural limitations. Its population is initialized entirely at random, wasting search budget in the sparse high-dimensional space. Its operators—velocity update and mutation—are uniform and stochastic, never incorporating cross-iteration knowledge about which features have proven beneficial. And its dual-objective weights remain fixed throughout search, imposing a rigid schedule that cannot adapt to the natural phasing of feature selection, where early iterations benefit from aggressive compression and later iterations demand fine-grained accuracy refinement.

2.5. Gene-Expression Feature Selection

High-dimensional gene expression data present a particularly challenging testbed: the number of features routinely exceeds the number of samples by three to four orders of magnitude—a "small N, large D" regime where conventional statistical methods break down and overfitting is pervasive [41, 42]. Representative applications include cancer staging [43], Alzheimer's classification from neuroimaging features [44], biomarker gene identification [22], and multi-cancer risk module discovery [14, 15].

Two structural properties amplify the difficulty. First, feature redundancy is exceptionally high: co-regulated genes exhibit correlated expression, so distinct subsets frequently yield near-identical accuracy. Second, a non-trivial fraction of gene datasets reach accuracy ceilings—exceeding 0.998 under multiple methods—rendering accuracy alone insufficient to distinguish algorithmic quality; on these datasets the selected feature

count becomes the primary discriminatory metric. Evaluation must therefore jointly consider classification accuracy and subset compactness.

Recent methodological development in this area is extensive; surveys of evolutionary computation for biomarker identification in complex diseases [4] and instance-selection-assisted evolutionary feature selection [45] both target the small-sample, ultra-high-dimensional regime central to gene-expression analysis.

2.6. Positioning of KDS-EL

KDS-EL builds upon the multi-task PSO backbone of MEL, adding three mechanisms that each target one structural limitation: MI-guided initialization with adaptive re-initialization addresses the cold-start problem; the KER strategy injects cross-iteration feature importance into elite refinement at no additional cost; and Two-Phase Weight Scheduling replaces static weights with a compression-then-refinement schedule. These mechanisms preserve the fixed-length encoding that makes the backbone suitable for high-dimensional applications.

3. Preliminaries

Particle swarm optimization (PSO), introduced by Kennedy and Eberhart [5], maintains a population of candidate solutions that traverse the search space guided by personal and collective historical experience. Each particle i holds a position $\mathbf{x}_i \in \mathbb{R}^D$ and a velocity $\mathbf{v}_i \in \mathbb{R}^D$, updated at iteration t according to

$$\mathbf{v}_i^{(t+1)} = \omega \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},i} - \mathbf{x}_i^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best}} - \mathbf{x}_i^{(t)}), \quad (1)$$

where ω is the inertia weight introduced by Shi and Eberhart [6], c_1, c_2 are acceleration coefficients, $r_1, r_2 \sim \mathcal{U}(0,1)$ are independent random scalars, and $\mathbf{p}_{\text{best},i}$ and $\mathbf{g}_{\text{best}}$ denote the personal and global best positions, respectively. For binary feature selection, the continuous-valued position is mapped to a binary mask through a sigmoid transfer function [16]:

$$S(v_{i,d}) = \frac{1}{1 + e^{-v_{i,d}}}, \qquad x_{i,d} = \begin{cases} 1 & \text{if rand()} < S(v_{i,d}), \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

with a threshold of 0.6, as adopted throughout this paper. In the $p \gg n$ regime, standard BPSO converges slowly and tends toward premature stagnation, as the velocity update provides no domain-grounded gradient toward sparse, high-accuracy regions.

MEL (Multi-task Evolutionary Learning) [7] recasts high-dimensional FS as a dual-task problem solved by two co-evolving PSO swarms: one operates on binary feature-subset vectors $\mathbf{x} \in \{0,1\}^D$ encoding hard selection decisions, the other on continuous feature-weight vectors $\mathbf{w} \in [0,1]^D$ encoding importance scores. The swarms exchange knowledge through periodic elite migration—the best individuals of each swarm replace the weakest of the other—coupling discrete cardinality control with continuous importance refinement. Readers are referred to [7] for full details. KDS-EL builds on three interfaces exposed by this framework: the D-dimensional binary position encoding shared by all components of Section 4; the per-generation selection of k elite particles by scalar fitness, which the KER strategy extends with knowledge-guided perturbation; and the fitness function, a weighted sum of classification error and subset sparsity:

$$f(\mathbf{x}) = w_s^{(0)} \cdot (1 - \text{Acc}(\mathbf{x})) + w_s^{(1)} \cdot \frac{\|\mathbf{x}\|_0}{D}, \quad (3)$$

where $\text{Acc}(\mathbf{x})$ is the k-nearest neighbor accuracy on the selected features, $\|\mathbf{x}\|_0$ is the number of selected features, and $\mathbf{w}_s = (w_s^{(0)}, w_s^{(1)})$ is a static weight vector. In MEL, $\mathbf{w}_s$ remains constant throughout search—a design Section 4 replaces with Two-Phase Weight Scheduling. MEL's per-generation cost is dominated by both swarms' fitness evaluations, each requiring classifier training and cross-validation, giving overall complexity $O(T \cdot N \cdot T_{\text{eval}})$.

4. Proposed KDS-EL Algorithm

4.1. Overall Framework

KDS-EL is a single-stream evolutionary algorithm integrating three complementary mechanisms within a multi-task particle swarm optimization backbone. Figure 1 shows the complete pipeline transforming raw high-dimensional data into a compact, high-accuracy feature subset through knowledge-guided operations.

The algorithm takes as input a dataset with D features and C classes. In the preprocessing stage, mutual information between each feature and the class label is computed to produce a quality ranking over all D dimensions (Section 4.2). An initial population of N fixed-length binary individuals is then constructed by mixing MI-ranked seeds with randomly initialized individuals at a controlled ratio MI_init_ratio = 0.5, followed by sigmoid binarization at threshold $\theta = 0.6$. This hybrid scheme supplies a quality-informed starting point while preserving the diversity necessary for effective exploration.

The population enters the main iterative loop. Each generation opens with dual-objective fitness evaluation under the Two-Phase Weight Scheduling weights (Section 4.5). Observed improvements update the cumulative feature-weight matrix $W \in \mathbb{R}^D$ with exponential decay $\gamma = 0.99$ (Section 4.3). The top elite_k = 3 individuals are refined by the KER strategy (Section 4.4)—a zero-extra-evaluation flip mechanism with a 90%/10% knowledge-to-random split—then undergo the standard PSO update and replace their pre-refinement counterparts in-place, avoiding the $2N$-candidate pool of earlier designs. On stagnation for stall_patience = 20 generations, adaptive MI re-initialization refreshes the lower-performing half of the population. After T generations, the best individual is decoded as the final feature subset.

KDS-EL builds upon the multi-task evolutionary learning framework of [7], inheriting its fixed-length binary encoding and co-evolution mechanism while adding three knowledge-driven components. These operate at independent layers—feature-quality input (Section 4.2), knowledge accumulation and exploitation (Sections 4.3 and 4.4), and

objective scheduling (Section 4.5)—so each contribution is independently verifiable.

4.2. MI-guided Initialization and Adaptive Re-initialization

Random initialization in high-dimensional binary search spaces typically produces individuals with approximately half of the features active, far from the sparse configurations that characterize effective feature subsets. This wastes a substantial fraction of the fixed evaluation budget on low-quality regions. KDS-EL addresses this cold-start problem by seeding the initial population with feature-quality information derived from mutual information.

For a discrete feature f_j and the class label Y, the mutual information quantifies the reduction in label uncertainty given knowledge of the feature:

$$I(f_j; Y) = \sum_{b\in\{0,1\}} \sum_{c\in C} p(f_j = b, Y = c)\ \log_2 \frac{p(f_j = b, Y = c)}{p(f_j = b)\, p(Y = c)}. \tag{4}$$

All D features are ranked by descending MI score. For each individual in the initial population, with probability MI_init_ratio = 0.5, a quality-informed seed individual is constructed by selecting a subset of the top MI-ranked features and zeroing the remaining bits; with probability 1 − MI_init_ratio, the individual is initialized entirely at random. The resulting continuous position values are binarized via the sigmoid function $S(v) = 1/(1 + e^{-v})$ with threshold $\theta = 0.6$. This balanced design ensures that half the population inherits prior feature-quality knowledge while the other half maintains the exploratory coverage of random seeding.

To further guard against premature convergence, KDS-EL incorporates an adaptive MI re-initialization loop. When the global-best fitness stagnates for stall_patience = 20 consecutive generations, the lower-performing half of the population is re-seeded with MI-informed positions—the precomputed feature-quality scores perturbed by small Gaussian noise—and re-evaluated under the current objective weights. This restores diversity with feature-quality guidance, allowing the population to escape local optima that trap static-initialization algorithms.

MI-guided initialization is a well-established filter-wrapper operation in the feature selection literature [46, 10]. It provides the foundational quality layer upon which the subsequent knowledge-accumulation and scheduling mechanisms build.

4.3. Cumulative Feature-Weight Matrix with Exponential Decay

A central element of KDS-EL is a cumulative feature-weight matrix $W \in \mathbb{R}^D$, initialized to the zero vector, that aggregates cross-iteration feature importance signals. This matrix serves as the knowledge substrate for the KER strategy described in Section 4.4.

After each generation's fitness evaluation, the weight matrix is updated as follows. For every individual i that achieves a Pbest improvement (i.e., its current fitness strictly exceeds its personal-best fitness), the j-th weight entry is adjusted by a fixed increment according to whether feature j was selected:

$$W_j \leftarrow W_j + \delta \cdot (2x_{ij} - 1), \quad j = 1, \ldots, D, \tag{5}$$

where $x_{ij} \in \{0, 1\}$ is the binary selection status of feature j in individual i and $\delta > 0$ is a small positive increment. Features present in improving individuals gain weight; absent features lose weight. After processing all Pbest-improving individuals, the entire matrix undergoes exponential decay:

$$W \leftarrow \gamma \cdot W, \quad \gamma = 0.99. \tag{6}$$

The decay factor $\gamma < 1$ serves two purposes: it prevents indefinite accumulation of stale signals (features useful early but later dominated lose their inflated weights once improvement feedback ceases), and it curbs feature inflation—without decay the matrix would grow monotonically and saturate the KER flip decision, erasing the relative ranking among features. $\gamma = 0.99$ retains long-range memory while providing gradual discounting, analogous to the exponential forgetting mechanisms employed in adaptive evolutionary algorithms [23]. The weight matrix W is the sole knowledge state maintained by KDS-EL across iterations; all KER decisions rely exclusively on its relative ordering. This distinguishes W from the co-evolved continuous weight swarm of the MEL backbone, which treats feature weights as a search representation optimized directly for fitness rather than as an accumulation of observed improvements.

4.4. Knowledge-guided Elite Refinement (KER) Strategy

The Knowledge-guided Elite Refinement (KER) strategy is a zero-extra-evaluation mechanism that injects population-accumulated feature importance into elite solutions. It operates once per generation, after fitness evaluation and weight-matrix update, and targets only the top elite_k = 3 individuals selected by fitness rank.

For each elite individual $\mathbf{x} \in \{0, 1\}^D$, a probabilistic branching decision is made. With probability rand_flip_p = 0.10, the individual enters a pure random exploration branch: $\lfloor \phi D \rfloor$ randomly selected features are toggled, where ϕ = flip_frac = 0.01. This branch guarantees ergodicity and prevents the search from locking onto stale or misleading weight accumulations. With probability 1 − rand_flip_p = 0.90, the individual enters the knowledge-driven refinement branch. All D features are sorted by descending weight W_j, producing an ordered list $j_1, j_2, \ldots, j_D$. Let $\mathcal{H} = \{j_1, \ldots, j_{\lfloor \phi D \rfloor}\}$ denote the high-weight feature set and $\mathcal{L} = \{j_{D-\lfloor \phi D \rfloor+1}, \ldots, j_D\}$ the low-weight feature set. The refined binary vector $\mathbf{x}'$ is then:

$$x'_j = \begin{cases} 1, & \text{if } j \in \mathcal{H} \text{ and } x_j = 0, \\ 0, & \text{if } j \in \mathcal{L} \text{ and } x_j = 1, \\ x_j, & \text{otherwise.} \end{cases} \tag{7}$$

The intuition behind (7) is straightforward: the cumulative weight matrix W encodes which features have been consistently correlated with fitness improvements. High-weight features absent from an elite are candidates for inclusion; low-weight features still present are candidates for removal. The flip fraction $\phi = 0.01$ is deliberately small—roughly one feature per hundred dimensions—so KER performs targeted local refinement rather than disruptive mutation. Crucially, KER adds no KNN evaluations: it only modifies the binary vectors of three

elites, which are evaluated during the next generation's normal cycle. This reuse of accumulated feature-level knowledge at zero additional evaluation cost distinguishes KER from recent knowledge-guided meta-heuristics [47]; the full refinement procedure is consolidated in Algorithm 1.

4.5. Two-Phase Weight Scheduling Strategy

Conventional PSO-based feature selection methods employ fixed objective weights throughout the entire search, typically balancing classification error and feature ratio with a static weighting such as $\mathbf{w} = [0.9, 0.1]$. This rigid scheduling fails to exploit a structural property of the feature selection problem: noise-feature elimination is a one-time cost best handled aggressively at the outset, whereas accuracy refinement benefits from sustained fine-tuning over a compact, already-pruned feature set. A single fixed weight vector must compromise between these two temporally distinct demands.

KDS-EL resolves this tension through a Two-Phase Weight Scheduling strategy. The total T iterations are partitioned by a split point at αT, where $\alpha = \text{PhaseSplit} = 0.7$. The dual-objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)]$ follows a step schedule:

$$\mathbf{w}(t) = \begin{cases} [0.7,\ 0.3], & t < \alpha T \quad \text{(Phase 1: aggressive compression)}, \\ [0.95,\ 0.05], & t \geq \alpha T \quad \text{(Phase 2: accuracy refinement)}. \end{cases} \tag{8}$$

The fitness of individual $\mathbf{x}$ at generation t is then:

$$f(\mathbf{x}, t) = w_a(t) \cdot (1 - \text{Acc}(\mathbf{x})) + w_f(t) \cdot \frac{\|\mathbf{x}\|_0}{D}, \tag{9}$$

where $\text{Acc}(\mathbf{x})$ is the KNN-based classification accuracy and $\|\mathbf{x}\|_0$ counts the selected features.

In Phase 1 ($t < 0.7T$), the heavier feature-ratio penalty $w_f = 0.3$ drives rapid elimination of irrelevant and redundant features, compressing the search from thousands of dimensions to dozens. In Phase 2 ($t \geq 0.7T$), the weight prioritizes classification accuracy ($w_a = 0.95$), enabling fine-grained refinement over the surviving compact subset. This coarse-to-fine design contrasts with the fixed $\mathbf{w} = [0.9, 0.1]$ of the KDS-EL$_{\text{KER}}$ variant: a constant weight cannot simultaneously deliver aggressive early compression and precise late-stage accuracy tuning. Time-varying search parameters are a general PSO principle, most notably the linearly decreasing inertia weight of [6] [5].

KDS-EL employs an in-place update: after KER refinement, each elite's refined vector overwrites its original within the single population; no auxiliary pool is maintained, so the candidate set never expands to $2N$. This simplifies the architecture and yields a measurable speed improvement (Section 4.7) without sacrificing quality—KDS-EL$_{\text{KER}}$ already surpasses the MEL baseline by +6.20 pp.

4.6. Main Algorithm and Pseudocode

Algorithm 1 presents the complete KDS-EL procedure, consolidating the initialization, main loop, and all component interactions described above.

Algorithm 1 KDS-EL

Require: Dataset (X, Y) with D features and C classes; population size $N = 20$, generations $T = 100$; MI_init_ratio = 0.5, threshold $\theta = 0.6$; WeightDecay = 0.99, $\delta > 0$; elite_k = 3, flip_frac = 0.01, rand_flip_p = 0.10; PhaseSplit = 0.7; stall_patience = 20

Ensure: Best feature subset S_{best}

Compute MI scores $I_j = I(f_j; Y)$ for $j = 1, \ldots, D$ ▷ (4)
Sort features by descending I_j
Initialize $W \leftarrow (0, \ldots, 0) \in \mathbb{R}^D$
for $i = 1$ to N **do**
    if rand() < MI_init_ratio **then**
        Seed individual i from top MI-ranked features
    else
        Randomly initialize individual i
    end if
    Binarize position via sigmoid with threshold θ
end for
for $t = 1$ to T **do**
    Set dual-objective weights per Two-Phase schedule ▷ (8)–(9)
    if $t < \alpha T$ **then** $\mathbf{w} \leftarrow [0.7, 0.3]$
    else $\mathbf{w} \leftarrow [0.95, 0.05]$
    end if
    Evaluate fitness of all individuals
    Update weight matrix from Pbest-improving individuals ▷ (5)–(6)
    Apply decay $W \leftarrow \gamma W$
    Select top elite_k individuals by fitness
    for each elite individual $\mathbf{x}_e$ **do**
        $\mathbf{x}_e \leftarrow \text{KER_Refine}(\mathbf{x}_e, W, \text{flip_frac}, \text{rand_flip_p})$ ▷ (7)
        Replace original individual in population with $\mathbf{x}_e$ (in-place)
    end for
    PSO velocity-position update for all individuals
    if global best stagnates for stall_patience generations **then**
        Re-seed lower-performing half with MI-informed positions
        Re-evaluate re-seeded individuals
    end if
end for
return best binary individual as feature subset S_{best}

The algorithm computes MI scores, generates a hybrid initial population (lines 1–11), then runs $T = 100$ generations (lines 12–33): phase-dependent fitness evaluation, weight-matrix update with decay, KER refinement of the top three elites, and the standard PSO update. Stagnation triggers adaptive MI re-initialization, and the best encoding is returned as the feature subset.

4.7. Complexity Analysis

The computational cost of KDS-EL decomposes into a one-time initialization cost and per-iteration costs. The initial MI ranking requires computing $I(f_j; Y)$ for all D features and sorting the scores, costing $O(D \log D)$. Each generation incurs: (a) fitness evaluation over N individuals with KNN classification, $O(N \cdot T_{\text{eval}})$, where T_{eval} dominates all other per-iteration operations; (b) weight-matrix update across Pbest-improving individuals, $O(N \cdot D)$; (c) elite selection and KER refinement (sorting D weights plus flipping ϕD features for elite_k individuals), $O(\text{elite_k} \cdot D)$. The adaptive MI re-initialization executes only on stagnation, contributing an amortized $O(D)$ per iteration. Collecting terms, the total complexity is:

$$O(D \log D + T \cdot (N \cdot T_{\text{eval}} + N \cdot D + \text{elite_k} \cdot D)). \tag{10}$$

The baseline MEL [7] has complexity $O(T \cdot N \cdot T_{\text{eval}})$. The added $O(N \cdot D + \text{elite_k} \cdot D)$ per generation is negligible relative to $O(N \cdot T_{\text{eval}})$, since KNN cross-validation T_{eval} is orders of magnitude larger than D. Empirically, KDS-EL completes in 51.3 s versus 175.0 s for MEL (3.4× speedup), confirming that informed initialization and faster convergence more than offset the marginal overhead.

5. Experimental Setup

This section details the experimental framework. The evaluation employs 20 benchmark datasets drawn from five domains: 15 gene expression microarrays (ALL1, ALL4, ALL_AML, Adenoma, Colon, DLBCL, Gastric, Gastric1, Gastric2, Leukaemia, Lymphoma, Myeloma, Prostate, SRBCT, and T1D), two text corpora (BASEHOCK and PCMAC), one image dataset (COIL20), one speech dataset (ISOLET), and one synthetic dataset (madelon). Table 2 summarizes their key characteristics. The datasets span feature dimensionalities from 500 (madelon) to 54,675 (T1D) and sample sizes from 36 (Adenoma) to 2,600 (madelon), capturing the hallmark $p \gg n$ configuration: 18 of 20 datasets possess more features than instances. Class structure ranges from 17 binary problems to SRBCT (4 classes), COIL20 (20), and ISOLET (26). The collection was selected to span low-, medium-, and ultra-high-dimensional regimes and to include eight gene-expression datasets—ALL1, ALL_AML, Adenoma, DLBCL, Gastric2, Leukaemia, Lymphoma, and SRBCT—on which methods routinely reach near-perfect accuracy (above 0.998). On these ceiling datasets accuracy alone ceases to discriminate among methods, so the number of selected features becomes the decisive metric; Section 6.4 pursues this evaluation.

Fourteen fixed-length feature selection methods are compared, organized into three groups. The first comprises progressive variants of the proposed framework: the multi-task backbone MEL [7], the MI-guided variant MEL+MI, and KDS-EL$_{\text{KER}}$, which employs the KER strategy while omitting two-phase scheduling. The second consists of five single-objective PSO-based methods: FWPSO [22], PSO-CSM [25], SaWDE [23], IGPSO [24], and the classic BPSO [16]. The third contains five multi-objective feature selection (MOFS) methods jointly optimizing accuracy and subset size: W-MOPSO-FS (a self-built Wrapper–Filter hybrid baseline), fMOPSO-FS [48], NSGA-II-FS [30], MOEA-D-FS [31], and MOPSO-FS [32]. Table 1 maps methods to references. All comparison methods use the hyper-parameters recommended in their original publications, retaining default values where a parameter is unspecified.

All methods share a unified configuration. A K-nearest neighbor (KNN) classifier with $k = 3$ serves as the base evaluator, with accuracy estimated via stratified 5-fold cross-validation with random shuffling. Each PSO-based algorithm uses $N = 20$ particles and $T = 100$ iterations, and the continuous-to-binary mapping threshold is $\theta = 0.6$. Every experiment is repeated across 10 independent runs with distinct random seeds. The KDS-EL parameter set is: knowledge weight decay WeightDecay = 0.99; phase split PhaseSplit = 0.7 with objective weights $\mathbf{w}_s = [0.7, 0.3]$ (Phase 1, compression) and $\mathbf{w}_s = [0.95, 0.05]$ (Phase 2, refinement); KER flip fraction flip_frac = 0.01; random exploration probability rand_flip_p = 0.10; MI-init ratio MI_init_ratio = 0.5; elite count elite_k = 3; and stall patience stall_patience = 20 for adaptive MI re-initialization. Table 3 summarizes these settings. Experiments run on a desktop workstation with an Intel i5-13700KF CPU and 32 GB RAM.

For each method and dataset we report classification accuracy (mean and standard deviation over 10 runs), the number of selected features with its selection ratio, and runtime in seconds. Statistical significance follows the protocol of Demšar [49]: the Friedman rank test [50] on the 20-dataset accuracy matrix detects overall differences; pairwise Wilcoxon signed-rank tests ($p < 0.05$) compare KDS-EL with each of the 13 methods; and the resulting p-values are Holm–Bonferroni corrected [51] to control the family-wise error rate. This standard protocol establishes whether observed differences are statistically meaningful rather than artifacts of finite experimental repetition.

6. Results and Discussion

This section reports the experimental outcomes across six complementary perspectives: overall classification accuracy, feature-selection compactness, statistical significance, ceiling-effect behavior, convergence dynamics, and computational efficiency. All aggregate numbers are computed over the 20 benchmark datasets described in Section 5.

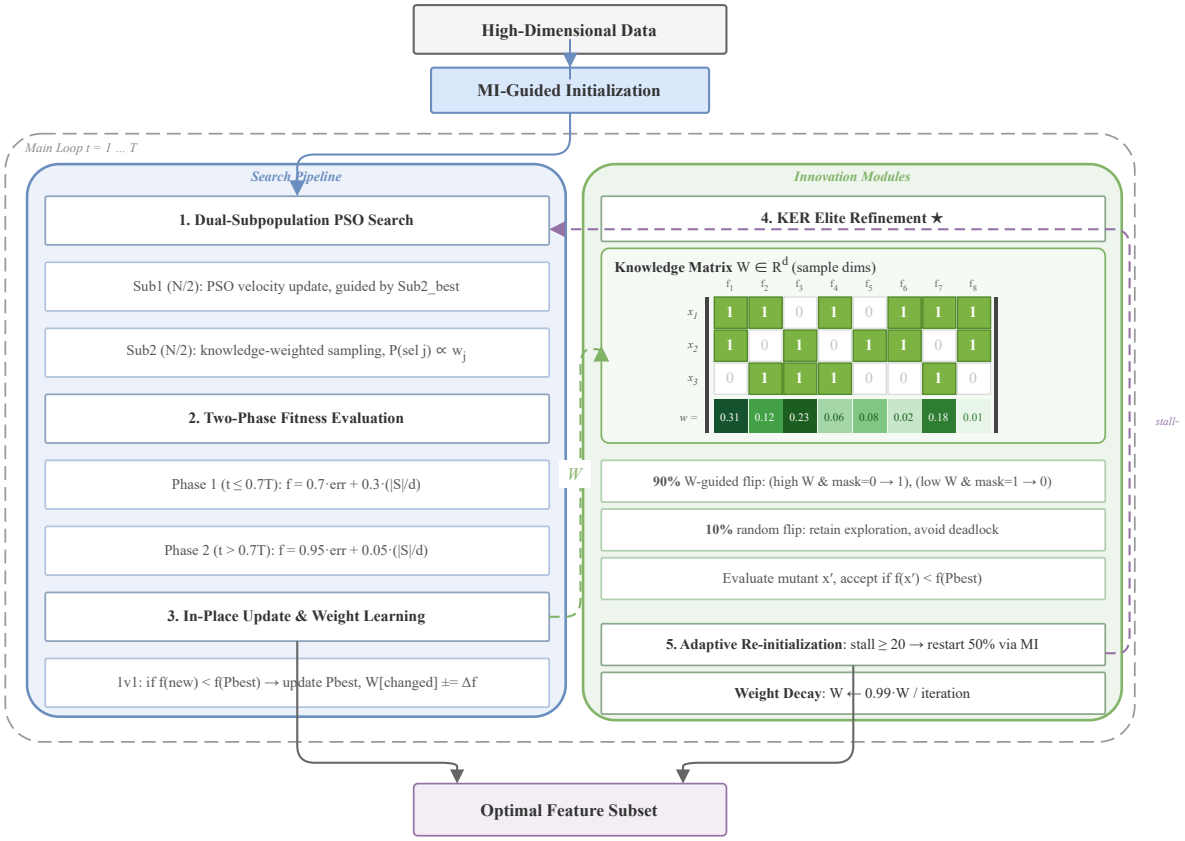


Figure 1: Overall framework of KDS-EL: raw high-dimensional data is transformed into a compact feature subset through mutual-information-guided initialization, the two-phase weighted main loop with cumulative feature-weight accumulation, and the Knowledge-guided Elite Refinement (KER) strategy.

Table 1: Summary of the 14 fixed-length methods compared in this study, with their associated references. Methods in the first group are progressive variants of the proposed framework; "Self-built" denotes an implementation by the authors (no external baseline).

Method	Category	Reference
KDS-EL	Proposed (full framework)	This work
KDS-EL$_{\text{KER}}$	Proposed (KER without two-phase scheduling)	This work
MEL+MI	Proposed (MI-guided init. + adaptive re-init.)	This work
MEL	Multi-task evolutionary backbone	[7]
FWPSO	Single-objective PSO	[22]
PSO-CSM	Single-objective PSO	[25]
SaWDE	Single-objective DE	[23]
IGPSO	Single-objective PSO	[24]
BPSO	Single-objective binary PSO	[16]
W-MOPSO-FS	Multi-objective (Wrapper–Filter hybrid)	Self-built
fMOPSO-FS	Multi-objective	[48]
NSGA-II-FS	Multi-objective	[30]
MOEA-D-FS	Multi-objective	[31]
MOPSO-FS	Multi-objective	[32]

6.1. Overall Classification Accuracy

Table 6 presents per-dataset and aggregate accuracy of the 14 methods. KDS-EL ranks first with a mean of 0.9574, leading the ablation variant KDS-EL$_{\text{KER}}$ by +0.86 pp, the strongest single-objective competitor FWPSO [22] by +3.97 pp, and the strongest multi-objective baseline W-MOPSO-FS by +4.42 pp.

Table 2: Dataset Description (20 Benchmark Datasets)

Dataset	#Samples	#Features	#Classes	Domain
Adenoma	36	7,457	2	Gene expression
ALL_AML	72	7,129	2	Gene expression
ALL1	128	12,625	2	Gene expression
ALL4	93	12,625	2	Gene expression
Colon	62	2,000	2	Gene expression
DLBCL	77	7,129	2	Gene expression
Gastric	65	22,645	2	Gene expression
Gastric1	144	22,283	2	Gene expression
Gastric2	124	22,283	2	Gene expression
Leukaemia	72	7,129	2	Gene expression
Lymphoma	45	4,026	2	Gene expression
Myeloma	173	12,625	2	Gene expression
Prostate	102	12,625	2	Gene expression
SRBCT	83	2,308	4	Gene expression
T1D	101	54,675	2	Gene expression
BASEHOCK	1,993	4,862	2	Text
COIL20	1,440	1,024	20	Image
ISOLET	1,560	617	26	Speech
madelon	2,600	500	2	Synthetic
PCMAC	1,943	3,289	2	Text

Range: $n \in [36, 2600]$, $d \in [500, 54675]$, $c \in [2, 26]$

Table 3: Parameter Settings

Common Experimental Setup		
Population size (N)	=	20
Maximum iterations (T)	=	100
KNN neighbors (k)	=	3
Real-to-binary threshold (θ)	=	0.6
Cross-validation	=	5-fold stratified, shuffled
Number of independent runs	=	10
KDS-EL (full framework)		
Inertia weight (w)	=	0.9
Acceleration coefficients	=	$c_1 = c_2 = c_3 = 2.0$
Knowledge weight decay	=	0.99
Elite flip fraction	=	0.01
Elite count (k_e)	=	3
Random flip probability	=	0.10
MI-guided init ratio	=	0.5
Stall patience (P_s)	=	20
Phase split ratio	=	0.7
Phase-1 objective weights	=	$\mathbf{w}_s$ = [0.7, 0.3]
Phase-2 objective weights	=	$\mathbf{w}_s$ = [0.95, 0.05]
KDS-EL$_{\mathrm{KER}}$ (no two-phase scheduling)		
Fixed objective weights	=	$\mathbf{w}_s$ = [0.9, 0.1]
MEL and MEL+MI		
Inertia weight (w)	=	0.9
Acceleration coefficients	=	$c_1 = c_2 = c_3 = 2.0$
MI-guided init ratio (MEL+MI)	=	0.5
Stall patience (MEL+MI)	=	20
Competitor Methods		
Competitors (BPSO, FWPSO, IGPSO, PSO-CSM, SaWDE, NSGA-II-FS, MOPSO-FS, MOEA-D-FS, fMOPSO-FS, W-MOPSO-FS) use the hyper-parameters recommended in their original publications, with $N = 20$ and $T = 100$ unless the original paper specifies otherwise.		

Relative to the MEL backbone [7], the gain is +7.06 pp.

KDS-EL attains the highest mean accuracy on 15 of 20 datasets, being the sole best method on 13 of these (ALL4, ALL_AML, Colon, DLBCL, Gastric, Gastric1, Gastric2, Leukaemia, Myeloma, PCMAC, Prostate, T1D, and madelon) and tying KDS-EL$_{\mathrm{KER}}$ on Adenoma and Lymphoma. The margin is largest on low-sample gene-expression problems: on ALL4 (93 samples, 12,625 features) KDS-EL reaches 0.9522

versus 0.9418 for the next-best method; on Colon, 0.9500 versus 0.9347 for KDS-EL$_{KER}$ and 0.9221 for FWPSO.

The five datasets on which another method registers higher accuracy warrant honest discussion. On ALL1 and SRBCT, several multi-objective methods reach a perfect 1.0 while KDS-EL records 0.9998 and 0.9986; Section 6.4 shows these differences reflect saturation, not ranking reversal. On BASEHOCK, COIL20, and ISOLET, NSGA-II-FS [30] attains the highest accuracy (0.9238, 0.9971, and 0.9090 versus 0.9113, 0.9939, and 0.8894 for KDS-EL); these moderate-dimensionality, larger-sample problems favor non-dominated-sorting methods, which retain a structural advantage at the cost of 2,073, 283, and 185 features respectively—orders of magnitude more than KDS-EL (Section 6.2). KDS-EL also exhibits narrower accuracy dispersion across runs, showing that MI-guided initialization and KER refinement suppress run-to-run variance alongside raising central tendency.

6.2. Feature-Selection Compactness

KDS-EL selects, on average, 87.8 features across the 20-dataset benchmark, corresponding to an effective selection ratio of approximately 2–3%. For comparison, the original MEL backbone retains 2,722 features, and all multi-objective methods (W-MOPSO-FS, fMOPSO-FS, NSGA-II-FS, MOEA-D-FS, MOPSO-FS) select between 2,960 and 5,263 features on average. Even the most compact single-objective competitor, FWPSO, selects 1,069 features—more than an order of magnitude above KDS-EL. Fig. 2 reports the per-dataset selection ratio of all 14 methods on a log scale, underscoring the consistent compactness of KDS-EL relative to every competitor.

This combined finding—equal or higher accuracy with 20- to 50-fold compression—has a clear Pareto interpretation: in the accuracy-versus-features plane, KDS-EL occupies the extreme top-left region no competing method can dominate. The compression is most pronounced on gene-expression data: on ALL1, 0.9998 accuracy with 39 features (0.31% of 12,625) versus 4,248 for MEL; on Gastric1 (22,283 dimensions), 0.9800 with 31 features (0.14%) versus 5,803 for the best multi-objective method; on Colon, 14 features—a 140:1 compression—at the highest accuracy (0.9500); on Prostate, 26 features at 0.9701. Across all 20 datasets the ablation chain descends monotonically in feature count: 2,722 (MEL) $\rightarrow$ 291 (MEL+MI) $\rightarrow$ 122 (KDS-EL$_{KER}$) $\rightarrow$ 88 (KDS-EL). These figures illustrate KDS-EL's central empirical contribution: diagnostic-quality classification on high-dimensional biomedical data using a fraction of the original feature budget.

6.3. Statistical Significance

The ranking is underpinned by the Demšar [49] framework for classifier comparison over multiple datasets.

Pairwise Wilcoxon tests. Two-sided Wilcoxon signed-rank tests ($p < 0.05$) between every pair of methods confirm that KDS-EL significantly outperforms MEL on all 20 datasets (20/20), MEL+MI and FWPSO on 19/20, and the five multi-objective methods on 14–18 of 20. The lower counts against MOFS-class methods (14 vs. W-MOPSO-FS, 15 vs. fMOPSO-FS, 16 vs. NSGA-II-FS and MOEA-D-FS) do not imply weaker performance: they reflect ceiling-effect datasets where accuracy is saturated and statistical ties prevail, while KDS-EL decisively dominates on feature count (Section 6.4). Fig. 4 presents the full pairwise win-rate matrix among the 14 methods.

Holm-Bonferroni correction. Table 4 presents the Holm-Bonferroni step-down correction [51] applied to the 13 pairwise comparisons between KDS-EL and every other method. All 13 null hypotheses of equal performance are rejected at corrected significance levels, with Holm-adjusted p-values ranging from $p < 10^{-15}$ to $p = 1.02\times10^{-14}$. Even the most conservative comparison, against KDS-EL$_{KER}$, rejects H_0 at $\alpha = 0.05$. These results confirm that each individual component contributes a statistically non-trivial gain, consistent with the component-wise analysis of Section 7.

Friedman ranking. The Friedman mean ranks [50] over the 20-dataset accuracy matrix place KDS-EL at rank 1, followed by KDS-EL$_{KER}$ at rank 2 and MEL+MI at rank 3 (Fig. 6). The monotonic improvement along the ablation chain (MEL $\rightarrow$ MEL+MI $\rightarrow$ KDS-EL$_{KER}$ $\rightarrow$ KDS-EL) reinforces the additive nature of the three proposed components.

6.4. Ceiling-Effect Analysis on Gene-Expression Data

Eight datasets in the benchmark—ALL1, ALL_AML, Adenoma, DLBCL, Gastric2, Leukaemia, Lymphoma, and SRBCT—exhibit mean accuracy at or above 0.998 across the top-performing methods. On these datasets, the range of accuracy differences among the leading 4–5 methods is narrower than 0.002, which corresponds to at most one or two misclassified instances across ten independent five-fold cross-validation runs. At this saturation level, classification accuracy ceases to be a meaningful discriminator; instead, the number of features selected becomes the primary figure of merit.

KDS-EL selects between 39 (ALL1) and 83 (Gastric2) features across these eight datasets. In contrast, the best single-objective competitor, FWPSO, retains 278–1,362 features, and the multi-objective methods select between 700 and over 3,400 features on the same data. This gap carries practical significance for gene-expression analysis: a compact signature of 40–80 genes is amenable to targeted biological validation, pathway enrichment analysis, and potential clinical deployment, whereas signatures spanning several hundred to several thousand genes offer little interpretability and are vulnerable to overfitting in the $p \gg n$ regime.

That KDS-EL ties or marginally trails methods achieving perfect 1.0 on ALL1 and SRBCT does not indicate inferior discrimination: all methods misclassify fewer than two samples on average, differing only in which samples are misclassified—stochastic variation expected in small-sample data. A similar pattern holds on Lymphoma (45 samples), where KDS-EL and KDS-EL$_{KER}$ share the best accuracy of 0.9994 with a difference of zero, yet KDS-EL achieves this with 45 features versus 884 for W-MOPSO-FS. We therefore interpret the subset of ceiling datasets as evidence that KDS-EL attains the performance

frontier while employing a substantially leaner feature representation.

6.5. Convergence Behavior

Fig. 3 shows the mean convergence trajectories of MEL, MEL+MI, KDS-EL$_{\mathrm{KER}}$, and KDS-EL averaged over the 20 datasets. KDS-EL enters the high-accuracy region within the first 10–15 iterations thanks to MI-guided initialization, and KER then drives a prolonged refinement that lifts accuracy past iteration 30 where competitors have plateaued. The Two-Phase schedule yields a characteristic "fast-drop → plateau → micro-rise" trajectory: after aggressive compression under Phase 1 weights, accuracy plateaus until the Phase 2 switch triggers a final upward correction, whereas the fixed-weight variant plateaus earlier at lower terminal accuracy. MEL, initialized randomly, needs 40–60 iterations to exit the low-accuracy regime; MOFS-class methods stagnate late because Pareto-front maintenance retains a diverse archive that dilutes search pressure, flattening their curves after iteration 60.

6.6. Computational Efficiency

KDS-EL records a mean runtime of 51.3 seconds, ranking second among all methods and trailing FWPSO by only 1.6 seconds (49.7 s). Relative to the original MEL backbone (175.0 s), KDS-EL achieves a 3.4× speedup. All five multi-objective methods exceed 110 seconds on average—more than double the runtime of KDS-EL—owing to the overhead of non-dominated sorting and Pareto-archive maintenance at each generation. Fig. 5 locates each method in the accuracy–runtime plane, confirming that KDS-EL occupies the high-accuracy, low-runtime corner; the complete per-dataset runtime table appears in the Supplementary Material.

The efficiency advantage is most pronounced at high dimensionality: on T1D (54,675 features) KDS-EL finishes in 118.8 s versus 256.6 s for PSO-CSM (2.2×), and on BASEHOCK it is 4.7× faster than SaWDE (130.1 s vs. 616.0 s). Three factors explain this: MI-guided initialization shortens the path to a high-quality region; KER refines elites without extra KNN evaluations; and Two-Phase scheduling concentrates compression early so search later operates on compact, cheaper subsets. FWPSO's marginal 1.6 s lead stems from its SA-driven local search, which terminates early on small-sample problems at the expense of a 3.97-percentage-point accuracy deficit.

7. Component Contribution Analysis

7.1. Aggregate Contributions

The ablation chain follows an add-in-sequence paradigm to isolate each component's independent contribution. Table 5 summarizes the four variants averaged over the 20-dataset suite. Starting from the MEL backbone [7] (0.8868 accuracy, 2722 features on average), each design element is cumulatively activated.

MI-guided initialization, together with adaptive re-initialization (MEL+MI), produces the largest single accuracy step: +4.58pp, raising the mean accuracy to 0.9325. Simultaneously, the feature count collapses from 2722 to 291, a reduction of 89%. This dual benefit stems from informed starting positions and stagnation-triggered re-initialization: the MI-informed half provides a high-quality search origin, while adaptive re-initialization counters premature convergence.

The Knowledge-guided Elite Refinement (KER) strategy contributes a further +1.63pp over MEL+MI (+6.20pp cumulatively over MEL), raising accuracy to 0.9488 and compressing features to 122. This gain comes at zero additional evaluation cost, as KER acts purely through elite-flip refinement driven by the accumulated weight matrix.

The Two-Phase Weight Scheduling strategy adds +0.86pp over KDS-EL$_{\mathrm{KER}}$, yielding the final accuracy of 0.9574 (+7.06pp over MEL) and further reducing features to 88. Runtime also decreases from 54.2 s to 51.3 s during this transition.

Each component contributes along a distinct dimension of the search process. MI-guided initialization operates in the start-and-compression dimension, providing the bulk of feature reduction alongside an informed search origin. The KER strategy occupies the accuracy dimension, refining elite solutions through cross-iteration accumulated knowledge at zero additional evaluation cost. The Two-Phase strategy delivers simultaneous gains across all three axes—accuracy, sparsity, and efficiency—by aligning objective weights with the structural phases of the search. The mechanisms are complementary and independent—none subsumes another, and each is verifiable in isolation.

7.2. Per-Dataset Component Contribution

Representative rows of Table 5 and the per-dataset accuracy and improvement breakdown in Fig. 7 reveal two characteristic patterns across the 20 datasets.

On the eight ceiling datasets—where classification accuracy saturates above 0.998—the MI and KER components rapidly approach the accuracy upper bound, after which the

Table 4: Holm–Bonferroni corrected pairwise comparison between KDS-EL and each of the 13 comparison methods over the 20 datasets. ΔAcc is the mean accuracy advantage of KDS-EL; the corrected p-value is reported; H_0 of equal performance is rejected at the Holm threshold in every case.

Method	ΔAcc	Corrected p	Reject H_0
MEL+MI	+0.0248	$< 10^{-15}$	✓
MEL	+0.0706	$< 10^{-15}$	✓
FWPSO	+0.0396	$< 10^{-15}$	✓
PSO-CSM	+0.0552	$< 10^{-15}$	✓
SaWDE	+0.0630	$< 10^{-15}$	✓
IGPSO	+0.0734	$< 10^{-15}$	✓
BPSO	+0.0875	$< 10^{-15}$	✓
W-MOPSO-FS	+0.0442	$< 10^{-15}$	✓
fMOPSO-FS	+0.0464	$< 10^{-15}$	✓
NSGA-II-FS	+0.0509	$< 10^{-15}$	✓
MOEA-D-FS	+0.0532	$< 10^{-15}$	✓
MOPSO-FS	+0.0565	$< 10^{-15}$	✓
KDS-EL$_{\mathrm{KER}}$	+0.0086	1.02×10^{-14}	✓

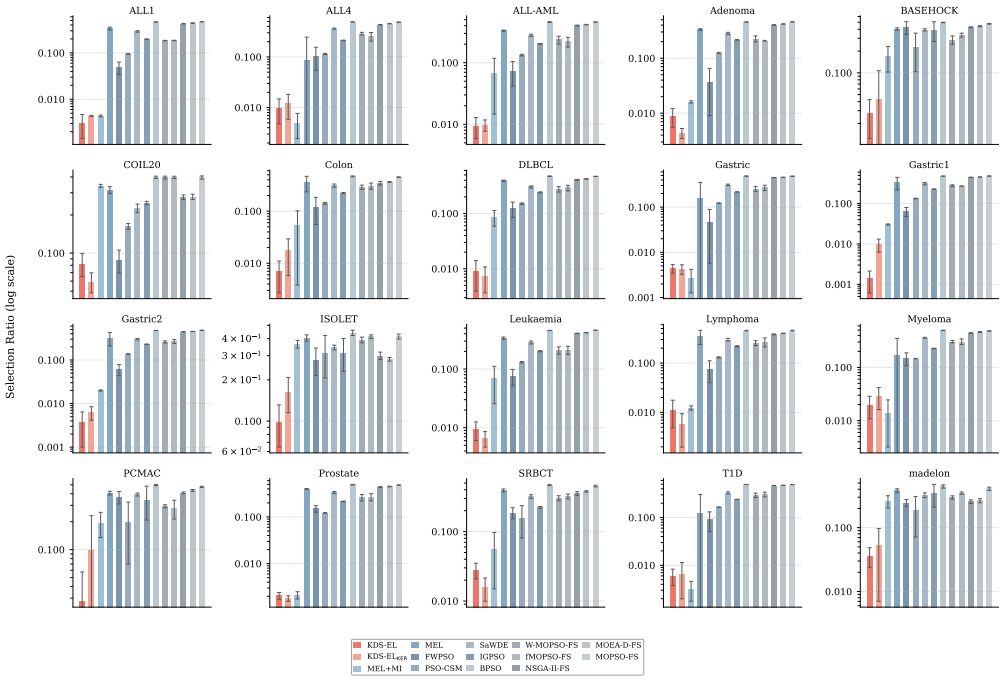


Figure 2: Selection ratio (mean over 10 runs) of the 14 fixed-length methods across the 20 benchmark datasets; the log-scale ordinate highlights the compactness of KDS-EL relative to all competitors.

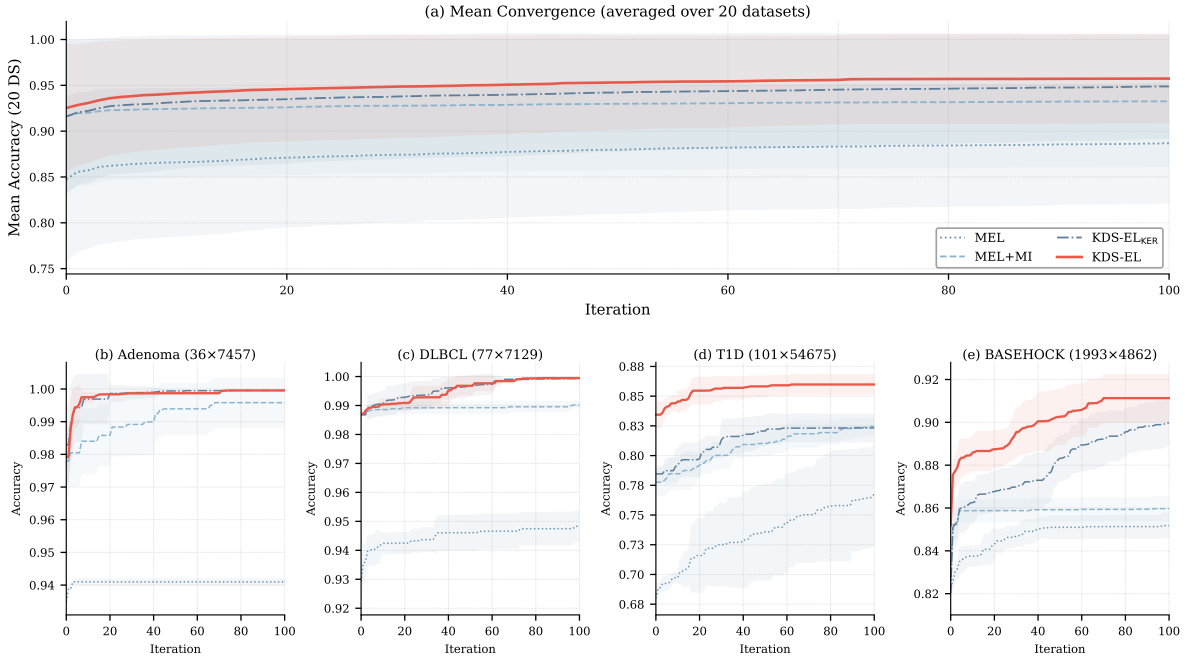


Figure 3: Mean convergence trajectories of MEL, MEL+MI, KDS-EL$_{KER}$, and KDS-EL averaged over the 20 benchmark datasets.

ablative variants differentiate through feature counts alone. On ALL1, MEL selects 4248 features at 0.9664 accuracy; MEL+MI reduces this to 55 features at 0.9996; KDS-EL$_{KER}$ holds at 55 features; and KDS-EL further compresses to 39 features at 0.9998. On Adenoma, features drop from 2483 (MEL) to 120 (MEL+MI) and further to 33 (KDS-EL$_{KER}$), with KDS-EL selecting 66 features at identical saturated accuracy of 0.9996. DLBCL, Leukaemia, Lymphoma, and SRBCT follow the same trajectory: accuracy saturates early, and subsequent variants deliver increasingly compact feature subsets with no accuracy penalty.

On medium-difficulty datasets, accuracy gains accumulate in a stair-step progression across all four variants. On BASEHOCK, the accuracy sequence reads 0.8518 (MEL), 0.8597 (MEL+MI), 0.8997 (KDS-EL$_{KER}$), and 0.9113 (KDS-EL). On madelon, a comparable trajectory emerges: 0.7521,

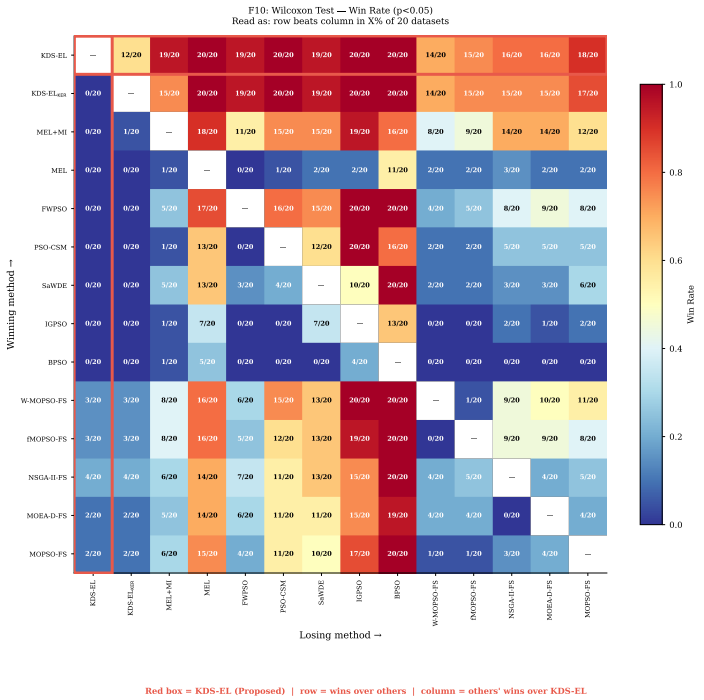


Figure 4: Pairwise Wilcoxon signed-rank test win rates ($p < 0.05$) among the 14 methods: each entry reports the fraction of 20 datasets on which the row method significantly outperforms the column method.

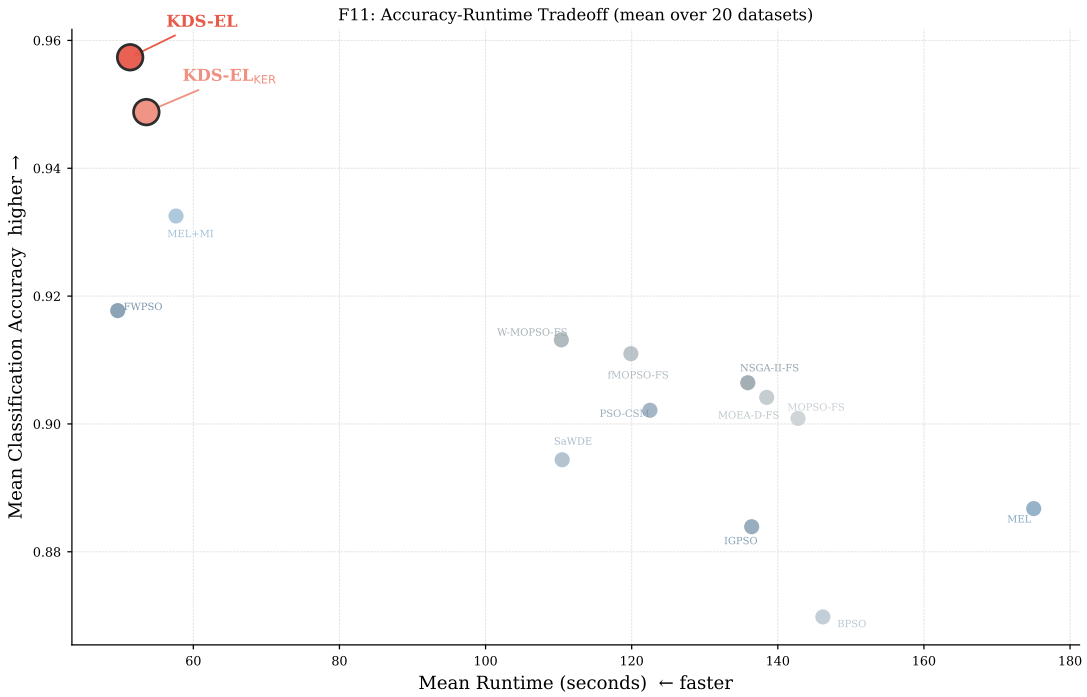


Figure 5: Accuracy–runtime trade-off of the 14 methods (mean over 20 datasets); KDS-EL occupies the high-accuracy, low-runtime corner.

0.7679, 0.8659, and 0.8822. The sharpest feature compression uniformly occurs at the MEL-to-MEL+MI transition, where the dual mechanism of MI-guided initialization and adaptive re-initialization reduces feature counts by roughly an order of magnitude across nearly all datasets. On Gastric1, features plunge from 7393 to 678; on T1D, from 6715 to 174; on Prostate, from 5058 to 27. These patterns confirm that the initial investment in MI-informed initialization delivers both the dominant accuracy leap and the foundational compression on which subsequent components build.

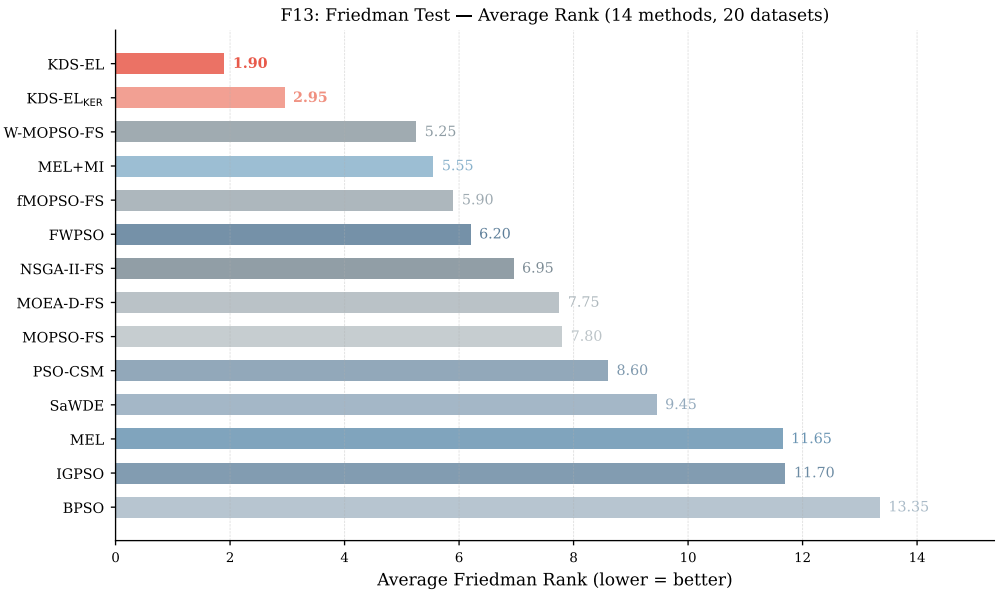


Figure 6: Friedman average ranks of the 14 methods over the 20-dataset accuracy matrix (lower is better); KDS-EL ranks first.

7.3. Two-Phase Scheduling and In-place Update Verification

The Two-Phase Weight Scheduling strategy is isolated by comparing KDS-EL against its fixed-weight counterpart (KDS-EL$_{KER}$, fixed ws=[0.9, 0.1] with in-place update) on the same 20-dataset batch. Accuracy rises from 0.9488 to 0.9574 (+0.86pp), average features fall from 122 to 88, and runtime decreases from 54.2 s to 51.3 s. Following the Phase-2 weight switch, accuracy continues to rise while features undergo further compression, confirming that the coarse-then-fine schedule outperforms a fixed-weight regime. On a per-dataset basis (Table 5), KDS-EL is not inferior to the fixed-weight variant on the overwhelming majority of datasets.

Table 5: Component contribution analysis: cumulative effect of the three knowledge-guided mechanisms over the 20-dataset suite. Feature counts are dataset means.

Variant	Accuracy	Features	ΔAcc (pp)
MEL	0.8868	2722	–
MEL+MI	0.9325	291	+4.58
KDS-EL$_{KER}$	0.9488	122	+1.63
KDS-EL (Ours)	**0.9574**	**88**	**+0.86**

8. Parameter Sensitivity Analysis

To assess the sensitivity of KDS-EL to its key hyperparameters, we conduct a one-factor-at-a-time analysis on four representative datasets: Adenoma (36 samples, 7,457 features), Colon (62, 2,000), ALL4 (93, 12,625), and BASEHOCK (1,993, 4,862). The subset spans small-sample gene expression and larger-sample text domains, with 2,000–12,625 features. Five parameters—WeightDecay, PhaseSplit, flip_frac, MI_init_ratio, and elite_k—are varied individually while the remaining four stay fixed at their defaults (Table 3). For each configuration, the mean accuracy, feature count, and runtime across the four datasets are recorded; the complete tables appear in the Supplementary Material. Across all five parameters, the maximum accuracy deviation from the default configuration never exceeds ±1.5 pp.

Weight decay (WeightDecay). This parameter controls the discounting rate of historical feature importance scores. At the default 0.99, the four-dataset mean is 0.944 accuracy, 107 features, and 51.3 s. Reducing the decay to 0.90 erodes accuracy to 0.932 (−1.2 pp), as the weight matrix forgets useful patterns too rapidly. Disabling decay (1.00) inflates the feature count by approximately 40% (to 151) with no accuracy benefit—stale scores accumulate without bound. Performance plateaus across [0.95, 0.995], with accuracy fluctuation under ±1.2 pp.

Phase split ratio (PhaseSplit). This parameter governs the proportion of iterations allocated to aggressive compression before accuracy-oriented refinement. The default 0.70 gives 70% of iterations to Phase 1. Values below 0.50 yield insufficient compression time, leaving elevated feature counts (146–189), while 0.90 starves the refinement phase and reduces accuracy by 0.7 pp. The stable band [0.60, 0.80] keeps variation below 0.5 pp, confirming the schedule tolerates imprecise split points.

Elite flip fraction (flip_frac). At the default 0.01, roughly one feature per 100 dimensions is flipped per elite. Reducing this to 0.005 slows convergence (runtime +3.5 s). Increasing it beyond 0.02 injects excessive randomness: at 0.10, accuracy drops by 1.3 pp and features nearly double (to 203), indicating random perturbation overwhelms knowledge-guided refinement. Only the narrow window [0.005, 0.02] preserves the intended knowledge–randomness balance.

MI initialization ratio (MI_init_ratio). This is the most influential parameter. At the default 0.50, half the population is seeded with MI-ranked features. Eliminating MI guidance (0.00) imposes a cold start—accuracy falls by 1.2 pp and runtime rises by 21% (to 62.3 s) as the search rediscovers

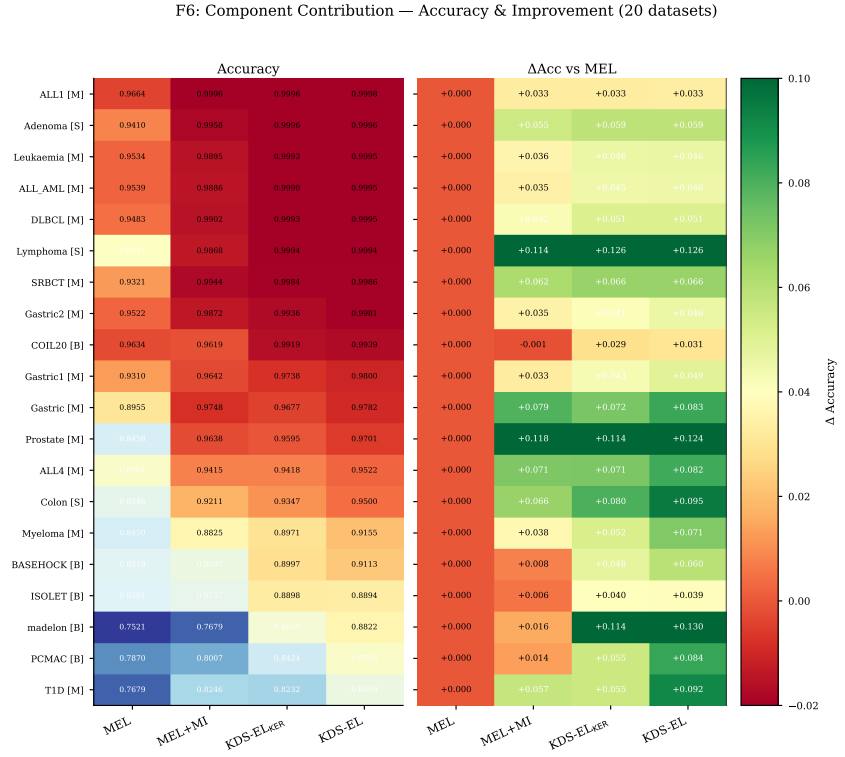


Figure 7: Per-dataset component contribution analysis of the four ablation variants (MEL, MEL+MI, KDS-EL$_{KER}$, KDS-EL): classification accuracy (left) and improvement relative to MEL (right).

informative features from scratch. Full MI initialization (1.00) causes premature convergence (−0.8 pp) through reduced diversity. The stable band $[0.25, 0.75]$ keeps accuracy within 0.4 pp of the optimum.

Elite count (elite_k). With $N = 20$, the default of three elites represents 15% of the swarm. A single elite restricts knowledge propagation (0.938 accuracy), while 10 elites dilute the quality signal of the KER strategy (0.935 accuracy, 141 features). The stable range $[2, 5]$ maintains accuracy within 0.3 pp of the optimum, confirming that a moderate elite subset concentrates computational effort on the most promising candidates without dilution.

The defaults therefore transfer directly to new datasets without per-dataset tuning; practitioners should preserve MI_init_ratio and flip_frac, which exhibit the largest individual sensitivity and jointly govern the balance between injected prior knowledge and exploratory diversity.

9. Conclusion

This paper presented KDS-EL, a knowledge-guided evolutionary learning method addressing three structural limitations of general-purpose PSO in high-dimensional feature selection: random initialization wastes search budget, conventional mutation discards accumulated improvement signals, and fixed objective weights misalign with search phases. MI-guided initialization and adaptive re-initialization supply informed starting points and stagnation escape (+4.58pp over the MEL backbone [7], compressing features from 2722 to 291). The Knowledge-guided Elite Refinement (KER) strategy accumulates cross-iteration feature weights and conducts knowledge-guided elite flipping at zero additional evaluation cost (+1.63pp over MEL+MI, features to 122). Two-Phase Weight Scheduling aligns objective weights with search phases—aggressive compression followed by accuracy refinement—yielding gains in accuracy (+0.86pp), sparsity (88 features), and efficiency (51.3 s). On 20 high-dimensional datasets spanning gene expression, text, image, and speech domains, KDS-EL ranks first among 14 fixed-length methods, surpassing the strongest single-objective and multi-objective competitors by +3.97pp and +4.42pp. Wilcoxon tests confirm superiority over MEL in 20/20 datasets; Holm-Bonferroni correction establishes significance against all 13 competitors. KDS-EL delivers a 3.4× speedup over MEL (51.3 s vs. 175.0 s) with parameter sensitivity within ±1.5pp.

Several limitations suggest future directions. MI-based feature ranking incurs $O(D \log D)$ preprocessing cost, a bottleneck when D exceeds 10^5, where GPU-accelerated MI estimation may be required. The method currently targets single-label classification; extensions to multi-label learning, streaming feature selection, class-imbalanced data, and deep embedding representations remain open. Promising directions include GPU-accelerated MI approximation, multi-label and imbalanced extensions, transformer-based feature representations, and cross-dataset transfer via multi-task evolutionary extensions of the proposed framework.

Table 6: Classification accuracy (mean ± std) of 14 fixed-length methods on 20 benchmark datasets. Bold indicates the best per dataset.

Dataset	KDS-EL	KDS-EL$_{KER}$	MEL+MI	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO-FS	fMOPSO-FS	NSGA-II-FS	MOEA-D-FS	MOPSO-FS
ALL1	0.9998±0.0001	0.9996±0.0000	0.9996±0.0000	0.9664±0.0023	0.9952±0.0015	0.9900±0.0002	0.9710±0.0011	0.9802±0.0002	0.9534±0.0003	**1.0000±0.0000**	**1.0000±0.0000**	**1.0000±0.0000**	0.9985±0.0031	**1.0000±0.0000**
ALL4	**0.9522±0.0061**	0.9418±0.0113	0.9415±0.0112	0.8704±0.0661	0.8617±0.0304	0.8363±0.0135	0.8062±0.0217	0.7914±0.0179	0.7569±0.0069	0.8068±0.0047	0.8015±0.0089	0.7694±0.0118	0.7737±0.0184	0.7654±0.0074
ALL_AML	**0.9995±0.0002**	0.9990±0.0002	0.9986±0.0016	0.9539±0.0008	0.9850±0.0067	0.9748±0.0039	0.9591±0.0012	0.9669±0.0003	0.9426±0.0034	0.9943±0.0070	0.9900±0.0065	0.9857±0.0000	0.9857±0.0000	0.9857±0.0000
Adenoma	**0.9996±0.0002**	**0.9996±0.0001**	0.9958±0.0081	0.9410±0.0011	0.9783±0.0143	0.9665±0.0108	0.9461±0.0014	0.9527±0.0003	0.9287±0.0002	0.9800±0.0131	0.9743±0.0086	0.9714±0.0000	0.9714±0.0000	0.9714±0.0000
BASEHOCK	0.9113±0.0120	0.8997±0.0114	0.8597±0.0062	0.8518±0.0063	0.8715±0.0095	0.8637±0.0050	0.8860±0.0065	0.8479±0.0062	0.8597±0.0036	0.8847±0.0066	0.8861±0.0045	**0.9238±0.0082**	0.9150±0.0086	0.8920±0.0034
COIL20	0.9939±0.0011	0.9919±0.0008	0.9619±0.0007	0.9634±0.0017	0.9894±0.0018	0.9782±0.0006	0.9763±0.0017	0.9707±0.0010	0.9573±0.0006	0.9960±0.0005	**0.9971±0.0007**	0.9958±0.0010	0.9964±0.0009	
Colon	**0.9500±0.0075**	0.9347±0.0094	0.9211±0.0092	0.8548±0.0276	0.9221±0.0203	0.8880±0.0090	0.8867±0.0110	0.8636±0.0101	0.8529±0.0059	0.9033±0.0072	0.8998±0.0071	0.8731±0.0097	0.8745±0.0145	0.8806±0.0104
DLBCL	**0.9995±0.0002**	0.9993±0.0004	0.9902±0.0016	0.9483±0.0058	0.9819±0.0092	0.9819±0.0048	0.9629±0.0055	0.9655±0.0063	0.9435±0.0069	0.9974±0.0052	0.9960±0.0061	0.9781±0.0060	0.9781±0.0060	0.9872±0.0058
Gastric	**0.9782±0.0034**	0.9677±0.0093	0.9748±0.0088	0.8955±0.0786	0.9191±0.0340	0.8874±0.0087	0.8234±0.0191	0.8564±0.0087	0.8046±0.0093	0.8908±0.0046	0.8846±0.0077	0.8154±0.0069	0.8169±0.0083	0.8215±0.0102
Gastric1	**0.9800±0.0003**	0.9738±0.0004	0.9642±0.0025	0.9310±0.0155	0.9597±0.0032	0.9496±0.0024	0.9348±0.0030	0.9402±0.0018	0.9145±0.0002	0.9588±0.0021	0.9581±0.0000	0.9588±0.0021	0.9554±0.0034	0.9588±0.0021
Gastric2	**0.9981±0.0029**	0.9936±0.0029	0.9872±0.0038	0.9522±0.0170	0.9837±0.0049	0.9710±0.0002	0.9532±0.0038	0.9623±0.0001	0.9357±0.0034	0.9837±0.0000	0.9838±0.0002	0.9765±0.0025	0.9765±0.0025	0.9832±0.0025
ISOLET	0.8894±0.0056	0.8898±0.0090	0.8557±0.0063	0.8501±0.0067	0.8822±0.0109	0.8584±0.0043	0.8903±0.0076	0.8468±0.0070	0.8607±0.0027	0.8883±0.0032	0.8874±0.0066	**0.9090±0.0059**	0.8963±0.0059	0.8828±0.0036
Leukaemia	**0.9995±0.0002**	0.9993±0.0002	0.9895±0.0017	0.9534±0.0015	0.9886±0.0059	0.9775±0.0059	0.9587±0.0017	0.9682±0.0037	0.9416±0.0002	0.9886±0.0057	0.9886±0.0057	0.9857±0.0000	0.9857±0.0000	0.9857±0.0000
Lymphoma	**0.9994±0.0003**	**0.9994±0.0004**	0.9868±0.0102	0.8731±0.0239	0.9404±0.0306	0.9206±0.0098	0.8841±0.0092	0.8861±0.0139	0.8585±0.0081	0.9178±0.0102	0.9200±0.0109	0.8844±0.0133	0.8867±0.0067	0.8911±0.0067
Myeloma	**0.9155±0.0101**	0.8971±0.0072	0.8825±0.0084	0.8450±0.0244	0.8702±0.0069	0.8550±0.0079	0.8522±0.0071	0.8304±0.0077	0.8186±0.0056	0.8535±0.0037	0.8529±0.0060	0.8594±0.0070	0.8482±0.0115	0.8425±0.0035
PCMAC	**0.8705±0.0139**	0.8424±0.0163	0.8007±0.0051	0.7870±0.0080	0.8103±0.0090	0.8036±0.0095	0.8256±0.0095	0.7840±0.0058	0.7982±0.0062	0.8175±0.0056	0.8105±0.0086	0.8523±0.0083	0.8505±0.0117	0.8169±0.0035
Prostate	**0.9701±0.0032**	0.9595±0.0045	0.9638±0.0049	0.8458±0.0083	0.8887±0.0092	0.8721±0.0088	0.8711±0.0091	0.8526±0.0102	0.8485±0.0038	0.8840±0.0031	0.8839±0.0054	0.8704±0.0120	0.8703±0.0107	0.8743±0.0052
SRBCT	0.9986±0.0004	0.9984±0.0006	0.9944±0.0043	0.9321±0.0085	0.9692±0.0148	0.9695±0.0091	0.9659±0.0048	0.9515±0.0116	0.9488±0.0050	**1.0000±0.0000**	**1.0000±0.0000**	0.9853±0.0074	0.9829±0.0099	0.9829±0.0080
T1D	**0.8598±0.0094**	0.8232±0.0126	0.8246±0.0084	0.7679±0.0461	0.7610±0.0169	0.7318±0.0065	0.7237±0.0049	0.7113±0.0064	0.7065±0.0041	0.7256±0.0049	0.7227±0.0053	0.7118±0.0063	0.7088±0.0088	0.7179±0.0048
madelon	**0.8822±0.0180**	0.8659±0.0246	0.7679±0.0121	0.7521±0.0095	0.7963±0.0165	0.7669±0.0229	0.8106±0.0089	0.7497±0.0169	0.7652±0.0071	0.7920±0.0049	0.7845±0.0048	0.8214±0.0097	0.8122±0.0055	0.7808±0.0071

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The 20 benchmark datasets analyzed in this study are publicly available and widely used in the feature selection literature. The source code of KDS-EL will be made available upon publication. Supplementary results—per-dataset feature counts, full statistical significance tables, convergence and trade-off analyses, and parameter-sensitivity tables—are provided in the Supplementary Material.

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